CISC 473: Deep Learning — Midterm Report

P22: Cross-Architecture Knowledge Distillation

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Abstract

Knowledge distillation (KD) compresses a large teacher network into a compact student, but most feature-level KD methods assume teacher and student share an architectural family. We study the harder cross-architecture case of a convolutional ResNet-50 teacher and a DeiT-Tiny vision-transformer student on CIFAR-100, where spatial feature maps must be related to patch-token sequences. To date we have implemented a deterministic CIFAR-100 pipeline (45k/5k/10k train/validation/test split), CIFAR-adapted ResNet-50 and DeiT-Tiny models, a shared training and evaluation engine, and resumable training scripts for the teacher and for the from-scratch student baseline. The ResNet-50 teacher reaches **[TODO: X.X]**% top-1 test accuracy **[TODO: over N seeds]**, compared with **[TODO: verify: 79.34]**% reported for the same architecture in [2,3], and the DeiT-Tiny student trained without a teacher reaches **[TODO: Y.Y]**%, which sets the lower bound that distillation must improve on. Next we will implement vanilla KD with temperature and weight ablations, measure layer-wise CKA between teacher and student, and add our proposed projected-feature alignment loss for the CNN-to-ViT setting.

Introduction and Problem Statement

Edge deployment favours small models, and vision transformers are increasingly the architecture of choice for compact students, yet the strongest teachers available for small-image tasks are often CNNs. Logit-only KD [1,3] transfers across architectures but discards the teacher's intermediate knowledge, while feature-level methods [2,4,5,7] were designed for CNN-to-CNN pairs whose feature maps are directly comparable. A CNN produces a $C \times H \times W$ spatial map and a ViT produces an $N \times D$ token sequence, so feature transfer between them needs an explicit alignment step.

Research question. *Does aligning projected intermediate CNN features with ViT patch tokens, using a CKA-based loss, give a DeiT-Tiny student accuracy gains on CIFAR-100 beyond logit-only KD, at unchanged inference cost, and is any gain accompanied by higher teacher–student CKA similarity?*

Scope changes since the proposal. **[TODO: Confirm the scope you are keeping. The repository README currently narrows the plan to ResNet-50 → DeiT-Tiny with vanilla KD and projected-CKA alignment, and marks CRD, DKD, ReviewKD, the ResNet-110 → MobileNetV2 pair, TinyImageNet and attention-map alignment as out of scope. The P22 objectives and the final rubric (≥2 baselines) still expect at least DKD or CRD, so either keep one of them or justify the change here.]**

Related Work (Updated)

Logit-based distillation. Hinton et al. [1] train the student to match the teacher's temperature-softened outputs, exposing the "dark knowledge" in non-target class probabilities. DKD [3] shows that the classical loss couples non-target knowledge to the teacher's target confidence and decouples the two terms, matching feature-based methods at logit-only cost. Because they act only on output distributions, both transfer unchanged across architectures, which makes vanilla KD our base objective and DKD the strongest logit-only reference.

Feature-based distillation. FitNets [5] introduced intermediate "hints" with a learnable regressor to reconcile channel widths; our projection module generalises this regressor from a channel mismatch to a map-versus-token mismatch. CRD [2] maximises a contrastive bound on teacher–student mutual information over pooled embeddings, ReviewKD [4] lets deeper student stages attend to shallower teacher stages, and RKD [7] transfers pairwise distance and angle structure, which is insensitive to representation dimensionality. All were developed and evaluated almost entirely on CNN pairs.

Cross-architecture distillation. DeiT [10] trains a ViT student from a CNN teacher (RegNetY) through a dedicated distillation token and finds that a convolutional teacher outperforms a transformer teacher, suggesting the CNN's inductive bias is itself useful knowledge; however, it distils logits only and modifies the student architecture with an extra token. Liu et al. [13] project CNN features into a transformer-like space with partially cross-attentive and group-wise linear projectors, and OFA-KD [14] projects intermediate features of heterogeneous students into the logit space and distils there, reporting gains on CIFAR-100 and ImageNet across CNN, ViT and MLP pairs. Our approach differs in aligning token-level representations directly with a CKA objective [6], adding no tokens to the student, and using CKA as a diagnostic of where transfer succeeds. **[TODO: Verify the details and arXiv IDs of [13] and [14] against the papers before submitting.]**

Method	Year	Key idea	Relation to our work
Vanilla KD [1]	2015	Match temperature-softened logits	Base loss term; first baseline
FitNets [5]	2015	L_2 hint loss with learnable regressor	Ancestor of our projection module
CKA [6]	2019	Invariant similarity of representations	Diagnostic and alignment loss
CRD [2]	2020	Contrastive teacher–student MI bound	Feature-level baseline (CNN pairs)
ReviewKD [4]	2021	Cross-stage review of teacher features	Motivates layer-pair ablation
DeiT [10]	2021	Distillation token, CNN teacher → ViT	Logit-only CNN→ViT reference
DKD [3]	2022	Decoupled target / non-target KD	Strongest logit-only baseline
OFA-KD [14]	2023	Distil heterogeneous features in logit space	Closest cross-architecture competitor

Methodology

Model Architecture

Teacher. ResNet-50 [8] from `torchvision`, randomly initialised (no pre-trained weights). For 32×32 inputs the 7×7 stride-2 stem is replaced by a 3×3 stride-1 convolution, the initial max-pool is removed, and the classifier outputs 100 classes. The `layer3` output is a $1024 \times 8 \times 8$ map. Parameters: **[TODO: confirm with `python -m src.models`; ≈23.7M]**.

Student. DeiT-Tiny [10] from `timm`, randomly initialised, with the standard 192-dimensional embedding, 12 blocks and 3 heads, but with input size 32 and patch size 4. This gives an 8×8 grid of 64 patch tokens plus a class token (sequence 65×192). Parameters: **[TODO: confirm; ≈5.4M]**.

Planned alignment module (not yet implemented). The teacher's 8×8 `layer3` map already matches the student's token grid, so it is flattened to 64 vectors and projected $\mathbb{R}^{1024} \to \mathbb{R}^{192}$ by a learnable 1×1 convolution. The total student objective is

$$\mathcal{L} = (1-\alpha)\,\mathcal{L}_{\mathrm{CE}} + \alpha\,\tau^2\,\mathrm{KL}(\sigma(z_T/\tau)\,\|\,\sigma(z_S/\tau)) + \beta\,\big(1 - \mathrm{CKA}(P(F_T),\,F_S)\big),$$

where P is the projection and F_S the student's patch tokens at the chosen block. The projection is discarded after training, so inference cost is unchanged.

Training Setup

Hyperparameter	Teacher (ResNet-50)	Student (DeiT-Tiny, scratch)
Optimiser	SGD, momentum 0.9	AdamW
Learning rate	0.1, cosine annealing	3×10^{-4}, 5-epoch linear warm-up then cosine
Weight decay	5×10^{-4}	0.05
Batch size	128	128
Epochs	100	100
Loss function	Cross-entropy	Cross-entropy
Data augmentation	Random crop (pad 4), horizontal flip	Same
Random seeds used	42 **[TODO: add seeds, e.g. 123, 999]**	42 **[TODO: add seeds]**

The checkpoint with the best validation top-1 accuracy is kept and evaluated once on the test set. Training is resumable from that checkpoint, restoring model, optimiser and learning-rate schedule state.

Dataset and Preprocessing

CIFAR-100 [12]: 60,000 32×32 colour images in 100 balanced classes. The official 50,000-image training set is split deterministically (split seed 42) into 45,000 training and 5,000 validation images; the official 10,000-image test set is used only for final evaluation. Images are normalised with per-channel CIFAR-100 mean and standard deviation; training images additionally receive a random 32×32 crop after 4-pixel padding and a random horizontal flip, while validation and test images are not augmented. Classes are balanced, so no re-weighting is applied.

Evaluation protocol. Top-1 and top-5 accuracy on the test set, reported as mean ± std over 3 seeds, together with parameter count and training time.

Preliminary Results

Baseline Reproduction

Target: Match the published number within 1–2% before proceeding to your proposed modification.

Model	Top-1 (%)	Top-5 (%)	Params
ResNet-50 teacher, published [2,3]	**[TODO: verify: 79.34]**	–	**[TODO:]**
ResNet-50 teacher, ours	**[TODO:]** ± **[TODO:]**	**[TODO:]** ± **[TODO:]**	**[TODO:]**
DeiT-Tiny student, scratch (ours)	**[TODO:]** ± **[TODO:]**	**[TODO:]** ± **[TODO:]**	**[TODO:]**
DeiT-Tiny + vanilla KD (early)	**[TODO: if available]**	**[TODO:]**	**[TODO:]**

Table 1: Preliminary results on CIFAR-100 test set. All results are mean ± std over **[TODO: 3]** seeds.

Qualitative Analysis

[TODO: Plot from teacher_metrics.csv and student metrics: train loss and validation top-1 per epoch. Remove duplicate epoch rows left by resumed runs first.]

Figure 1: Training loss and validation top-1 accuracy per epoch for the ResNet-50 teacher and the from-scratch DeiT-Tiny student on CIFAR-100.

Discussion

[TODO: Interpret your numbers. Points worth covering: (1) how close the teacher is to the published 79%; (2) the size of the teacher–student gap, which is the headroom KD can recover; (3) how much faster or slower the ViT converges than the CNN, since ViTs trained from scratch on small datasets are known to underfit without a CNN's locality prior; (4) any overfitting seen as a gap between train and validation accuracy.]

Challenges and Adaptations

Compute interruptions. Training uses the Google Colab free tier, whose sessions end after roughly 12 hours or on inactivity, deleting anything stored on the runtime's local disk. We moved checkpoints and metric logs to Google Drive and added a `--resume` option that restores model, optimiser and learning-rate schedule from the last best checkpoint, which let the 100-epoch teacher run complete across several sessions.

Correct resumption. An early version restarted the cosine schedule from the wrong epoch after a resume and mixed zero- and one-based epoch numbers; we fixed both so a resumed run follows the same learning-rate curve as an uninterrupted one. Because only the best checkpoint is saved, resuming can repeat a few epochs and append duplicate rows to the metrics log; these are removed before plotting.

Feature-shape compatibility. Choosing patch size 4 for DeiT-Tiny gives an 8×8 token grid that matches ResNet-50's `layer3` map, so the alignment module needs no spatial interpolation for that layer pair. We verify these shapes with forward hooks in `src/models.py`.

Timeline. **[TODO: State whether you are behind the proposal: the proposal scheduled vanilla KD (Week 3), CRD (Week 4), DKD and the cross-architecture setup (Week 5) and ReviewKD and CKA (Week 6). So far the teacher and scratch-student baselines are done. Explain why (Colab limits, repo rebuild) and the reduced or re-ordered plan below.]**

Group size. **[TODO: The guide requires groups of 3; explain or remove.]**

Updated Week 7–10 Plan

Week	Tasks	Responsible
7	Vanilla KD (ResNet-50 $\rightarrow$ DeiT-Tiny); $\tau \in \{1, 2, 4, 8, 16\}$ and α ablation; layer-wise CKA tooling	**[TODO: Name]**
8	Projected-CKA alignment module; β and layer-pair ablations; **[TODO: DKD baseline if kept]**	**[TODO: Name]**
9	3-seed reruns of selected configurations; CKA heatmaps before/after KD; accuracy vs. FLOPs analysis	**[TODO: Name]**
10	Final report, presentation, single-command reproduction and README **(Final due)**	All

Preliminary Novel Contribution (Optional Bonus)

[TODO: Leave empty unless you have evidence. A quick candidate: compute layer-wise CKA between the trained teacher and the scratch student and show which layers are already similar or dissimilar before any distillation, as a heatmap. That is a quantitative, not-yet-reported baseline for the CNN$\rightarrow$ViT CKA study.]

AI Tool Disclosure

[TODO: Required by the course guide. List the AI tools used and for what, e.g. "Claude was used to draft the structure and wording of this report from our proposal and code, and for debugging help; all experiments, results and analysis are our own and were verified by us." Edit to match what you actually did.]

Contribution Declaration

Member	Contributions (this period)	%
Lee Xuan	**[TODO: e.g. Related-work update, teacher training runs, report writing]**	**[TODO:]**%
Fong Yee Xin	**[TODO: e.g. Data pipeline, models, training engine, resumable training scripts]**	**[TODO:]**%

References

[1] Hinton, G., Vinyals, O., and Dean, J., 2015. "Distilling the Knowledge in a Neural Network." *NeurIPS Deep Learning and Representation Learning Workshop*. arXiv:1503.02531.

[2] Tian, Y., Krishnan, D., and Isola, P., 2020. "Contrastive Representation Distillation." *ICLR*. arXiv:1910.10699.

[3] Zhao, B., Cui, Q., Song, R., Qiu, Y., and Liang, J., 2022. "Decoupled Knowledge Distillation." *CVPR*. arXiv:2203.08679.

[4] Chen, P., Liu, S., Zhao, H., and Jia, J., 2021. "Distilling Knowledge via Knowledge Review." *CVPR*. arXiv:2104.09044.

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[6] Kornblith, S., Norouzi, M., Lee, H., and Hinton, G., 2019. "Similarity of Neural Network Representations Revisited." *ICML*. arXiv:1905.00414.

[7] Park, W., Kim, D., Lu, Y., and Cho, M., 2019. "Relational Knowledge Distillation." *CVPR*. arXiv:1904.05068.

[8] He, K., Zhang, X., Ren, S., and Sun, J., 2016. "Deep Residual Learning for Image Recognition." *CVPR*. arXiv:1512.03385.

[9] Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., and Chen, L.-C., 2018. "MobileNetV2: Inverted Residuals and Linear Bottlenecks." *CVPR*. arXiv:1801.04381.

[10] Touvron, H., Cord, M., Douze, M., Massa, F., Sablayrolles, A., and Jégou, H., 2021. "Training Data-Efficient Image Transformers & Distillation through Attention." *ICML*. arXiv:2012.12877.

[11] Dosovitskiy, A., et al., 2021. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." *ICLR*. arXiv:2010.11929.

[12] Krizhevsky, A., 2009. "Learning Multiple Layers of Features from Tiny Images." Technical report, University of Toronto.

[13] Liu, Y., Cao, J., Li, B., Hu, W., Ding, J., and Li, L., 2022. "Cross-Architecture Knowledge Distillation." *ACCV*. arXiv:2207.05273. **[TODO: verify]**

[14] Hao, Z., Guo, J., Han, K., Tang, Y., Hu, H., Wang, Y., and Xu, C., 2023. "One-for-All: Bridge the Gap Between Heterogeneous Architectures in Knowledge Distillation." *NeurIPS*. arXiv:2310.19444. **[TODO: verify]**